Gloss, Stitch, Sign: Low-Resource Sign Language Generation for Turkish Sign Language with LLMs and a Community Dictionary

Anonymous ACL submission

Abstract

Sign language generation for under-resourced languages remains challenging due to limited parallel corpora. We present a modular Turkish-to-Turkish Sign Language (TİD) generation framework: LLMs translate Turkish into TİD glosses, two linguistically motivated modules (spatial agreement rewriting and context-aware resolution of out-of-vocabulary and polysemous glosses) turn the glosses into a pose sequence stitched from dictionary clips, and a 3D avatar renders the result. The framework is built on TİD Sözlük, a community-developed dictionary with 3,706 glossed example-sentence videos. We compare fine-tuned open LLMs (Gemma 3, Qwen 3) with prompted closed LLMs (Gemini 3 Flash, GPT-5.4) on 371 sentences. The larger fine-tuned open models outperform the closed models on BLEU, chrF and ROUGE, and semantic similarity differs little. Both modules give small but significant improvements in hand-trajectory alignment for most systems, with context-aware resolution producing the larger effect. In a study with Deaf and expert TİD signers, expert signers rated avatar videos built from LLM glosses on par with videos built from gold glosses and recovered part of the meaning in blind translation; the remaining gap to real signing lies in sign synthesis and non-manual signals.

1 Introduction

Sign languages are natural languages with their own morphology, syntax, and spatial reference systems (Sandler and Lillo-Martin, 2006). Despite progress for American Sign Language (ASL) (Huenerfauth, 2008) and German Sign Language (DGS) (Saunders et al., 2020), Turkish Sign Language (TİD)[1] remains *under-represented* in computational research. Resources such as BosphorusSign (Camgöz et al., 2016), AUTSL (Sincan and Keles, 2020), and the community-developed TİD Sözlük (Makaroğlu and Dikyuva, 2017) exist, but prior TİD work has focused mainly on isolated sign recognition (Özdemir et al., 2020) and pose estimation (Keleş et al., 2023), with limited rule-based and statistical machine translation (Kayahan and Güngör, 2019). TİD has also not been systematically incorporated into education or health services, leaving Deaf signers underserved (Kemaloğlu and Kemaloğlu, 2012). Sign language generation can help by translating written language into sign language video (Saunders et al., 2020; Walsh et al., 2024b), but current methods were developed mainly for higher-resource sign languages. TİD is not signed Turkish: it has its own grammar and spatial mechanisms and is historically unrelated to the most studied American and European sign languages (Zeshan, 2002, 2003; Zwitserlood et al., 2012). Because TİD has little parallel data, we study a gloss-to-pose route with explicit TİD-specific rules, beginning with LLMs for Turkish-to-TİD glossing. We organize our investigation around:

- **RQ1.** How well do current LLMs translate Turkish into TİD-conventional glosses, and does parameter-level fine-tuning of open LLMs outperform in-context prompting of closed LLMs on this task?
- **RQ2.** Does explicit modeling of TİD-specific linguistic structure, namely spatial agreement and context-aware resolution of out-of-vocabulary (OOV) and polysemous glosses, improve the pose sequences stitched from dictionary clips, as measured by module ablations?

To address these questions, we build on TİD Sözlük (Makaroğlu and Dikyuva, 2017), a dictionary

[1] TİD is the language of the Deaf community of Türkiye. Kemaloğlu and Kemaloğlu (2012) estimate the deaf population of Türkiye at 187,500–337,500, about 0.25%–0.45% of the 2012 population; the number of TİD signers is not known and is not the same as the number of deaf people.

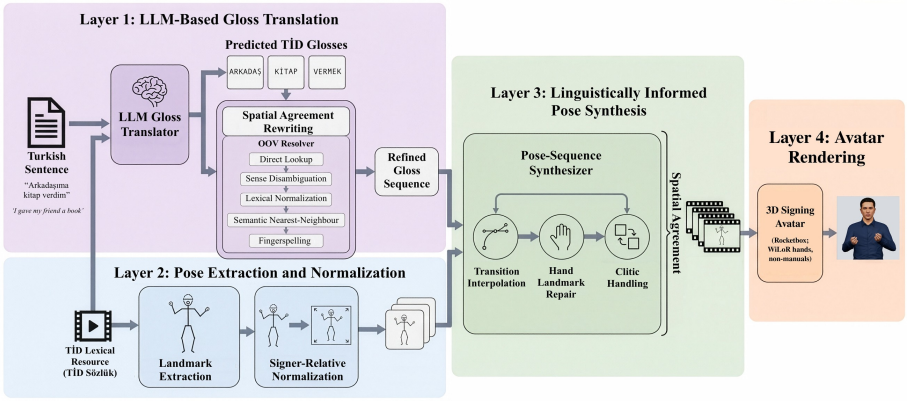


Figure 1: Architecture of the Turkish-text-to-TİD generation pipeline. Turkish input is converted into TİD glosses by fine-tuned or prompted LLMs. The predicted glosses are then passed through TİD-specific post-processing, including spatial agreement, OOV resolution, and dictionary lookup, before lexical sign clips are stitched into a pose sequence. The final stage renders the pose sequence as video with an existing 3D avatar.

resource developed with the Deaf community. Our pipeline has three stages (Figure 1): (i) fine-tuned or prompted LLMs translate Turkish into glosses; (ii) deterministic post-processing applies spatial agreement rewriting and context-aware OOV resolution and stitches dictionary clips into a pose sequence; and (iii) an existing 3D signing avatar renders the pose sequence. The first two stages are the contributions of this paper; the third is a visualization tool.

We build on existing tools: MediaPipe Holistic (Lugaresi et al., 2019) and WiLoR (Potamias et al., 2025) for pose and hand estimation, sentence embedding models (Reimers and Gurevych, 2019) for OOV resolution, and the Rocketbox avatar library (Gonzalez-Franco et al., 2020) for rendering. Our contribution is the TİD-specific pipeline that connects them and its evaluation with automatic metrics and native signers. The contributions of this work are:

1. A *linguistically motivated spatial agreement* module that, drawing on the descriptive grammar of TİD (Kubuş, 2008; Sevinç, 2006), inserts discourse loci and rewrites the movement and orientation of verbs at the pose level.
2. A *context-aware OOV resolution* pipeline that resolves predicted glosses through direct lookup, sense disambiguation, lexical normalization, semantic nearest-neighbor substitution, and fingerspelling, yielding the larger of the two module improvements in hand-trajectory alignment.
3. A systematic comparison of *fine-tuning open LLMs* (Gemma 3 and Qwen 3 at multiple scales) against *in-context prompting of closed LLMs* (Gemini 3 Flash and GPT-5.4) for the low-resource Turkish-to-TİD gloss task, with paired significance tests, and ablations of the two modules above.
4. A *diagnostic evaluation with TİD signers* (four Deaf signers and four expert signers, 25 test sentences), which locates the remaining errors in the pipeline.

2 Related Work

Sign Language Generation Early sign language generation systems relied on rule-based translation with avatar animation (Huenerfauth, 2008; Veale et al., 1998). Statistical and neural methods later replaced these pipelines, including pose-based autoregressive generation (Saunders et al., 2020; Walsh et al., 2024a) and video synthesis via GANs or diffusion models (Fang et al., 2023; Qi et al., 2025). Several works model articulators beyond the hands, addressing non-manual markers (Zhang et al., 2025), concept-motivated production for STEM (Sortino et al., 2026), and pragmatic articulation in classroom discourse (Imai et al., 2025b). More expressive body representations such as parametric 3D meshes have also been adopted (Pavlakos et al., 2019; Zuo et al., 2025). Related work learns sign representations (Wong et al., 2025; Gueuwou et al., 2025b; Jiang et al., 2024; Gueuwou et al., 2025a), uses phonological priors (Kezar et al., 2023, 2025b,a) and segmentation (Moryossef et al., 2023), and proposes evaluation metrics and benchmarks (Imai et al., 2025a; Inan et al., 2024). Dictionary-based production remains attractive for low-resource settings since it exploits existing lexical video, but

naive concatenation produces discontinuities. Sign-stitching methods (Walsh et al., 2024b) normalize sign clips and interpolate between boundaries. Our system follows this dictionary-stitching line while adapting it to the lexical conventions and pose-level constraints required for TİD generation. Neural text-to-pose and video generation methods (Saunders et al., 2020; Walsh et al., 2024a; Fang et al., 2023; Qi et al., 2025) learn this mapping from aligned sentence-level sign video and are trained on larger parallel corpora than the 3,706 TİD Sözlük example-sentence pairs available for TİD (Section 3), which are also our only source of evaluation data. Dictionary-based production needs only isolated signs, which are available for TİD. Existing stitching systems, however, do not change the movement of agreement verbs or choose among dictionary senses by sentence context; these are the two modules we add for TİD.

LLMs, Glossing, and Low-Resource Translation In-context machine translation with LLMs has recently been studied as a promising approach for low-resource MT, especially when prompts can incorporate dictionaries, grammar descriptions, or retrieved parallel examples (Tanzer et al., 2024b). A growing body of work applies them to sign language. Several efforts use LLMs as gloss-free or gloss-mediated translators (Wong et al., 2024; Gong et al., 2024; Chen et al., 2024; Fayyazsanavi et al., 2024), with Zhang et al. (2025) showing that GPT-4o supports English-to-ASL glossing under carefully designed prompting and Inan et al. (2025a) integrating multimodal sign processing directly into a pretrained LLM. Other work scales sign language translation (Zhang et al., 2024; Gueuwou et al., 2025a), revisits its evaluation (Tanzer et al., 2024a), and extends it to low-resource settings (Gueuwou et al., 2023; Inan et al., 2025b; Lea et al., 2026). For TİD, we combine LLM-based gloss generation, comparing fine-tuned open LLMs against prompted closed LLMs, with post-processing rules derived from TİD Sözlük conventions and from the descriptive grammar of TİD.

3 Data and Preprocessing

3.1 Corpora

We build on the publicly available TİD Sözlük resource[2] (Makaroğlu and Dikyuva, 2017), a corpus-based dictionary of the 2,000 most frequent lemmas in a TİD corpus collected across 26 cities in Türkiye from 116 native signers, which gives it broad coverage of regional variation. The version we use lists 2,013 lemma entries with 3,790 senses; each sense has isolated-sign videos (one per variant) and an example-sentence video with a Turkish translation and gloss transcription. The example sentences are our parallel data: 3,706 senses have a gloss of at least two tokens, giving 3,706 Turkish–TİD pairs. Lexical items that cannot be resolved against TİD Sözlük are fingerspelled with the manual alphabet from İşaretçe.[3]

3.2 TİD Gloss Conventions

We inspected TİD Sözlük sentence labels and use the following conventions for prompting and post-processing. TİD glosses are all-caps Turkish lemma/root forms; compounds are marked with ^, especially signs such as BILMEK^DEĞIL (KNOW^NOT)[4]; repetition and continuous aspect are represented by repeated glosses, and parenthesized tokens indicate optional material and are removed or separated during parsing. The corpus labels do not encode how signs are directed in the space, so the LLM is trained to produce corpus-style glosses; spatial marking is added later by the rewriting module described in Section 4.2.

3.3 Pose Extraction and Representation

We use MediaPipe Holistic (Lugaresi et al., 2019) to extract body and hand landmarks from TİD Sözlük videos, storing 33 pose landmarks and 21 landmarks per hand for each frame. We normalize pose coordinates relative to shoulder landmarks, apply temporal smoothing, validate pose completeness, and filter sequences with excessive missing skeletal points. The system then uses a signer-relative canonical frame to compensate for signer placement and camera angle variation. Each sign's temporal profile is trimmed using velocity and signing-space criteria to remove non-signing preparation and retraction.

[2] https://tidsozluk.aile.gov.tr/

[3] https://isaretce.com/, a publicly available TİD resource of 2,411 signs, of which we use only the alphabet. The letter clips are converted into wrist-relative canonical hand poses and placed into the active signer frame during synthesis; fingerspelling is the final fallback after lexical and semantic substitution (Section 4.3).

[4] We follow the sign-language linguistics convention of writing sign glosses in small caps to distinguish them from ordinary spoken or written words.

3.4 Training, Validation, and Test Splits

The 3,706 pairs serve both as supervision and for evaluation. We hold out 371 test sentences, used for gloss-level and pose-level evaluation, and split the remaining pairs (after removing a few duplicates of test items) at random into training (85%, 2,827 pairs) and validation (15%, 500 pairs) sets. Training and test sentences come from the same dictionary and signers, so the test set measures in-domain performance.

4 Our Framework

Our framework has three stages: gloss prediction from Turkish text, linguistically informed pose-sequence construction (spatial agreement rewriting, context-aware OOV resolution, and clip stitching), and avatar rendering.

4.1 LLM-Based Gloss Translation

The first stage translates Turkish sentences into TİD glosses. We evaluate fine-tuned open models, Gemma 3 (Gemma Team et al., 2025) at 4B, 12B, and 27B and Qwen 3 (Yang et al., 2025) at 4B, 8B, and 32B, against Gemini 3 Flash (Google DeepMind, 2025) and GPT-5.4 (OpenAI, 2026) under one-shot and ten-shot prompting.

Open models are fine-tuned with LoRA on the training split; hyperparameters, decoding settings and compute are in Appendix C (Table 4). Closed models are prompted only, with one or ten randomly selected training pairs as in-context examples; the system message is reproduced in Appendix B.

4.2 Spatial Agreement Rewriting

We follow the standard distinction between plain, agreement, and spatial verbs (Padden, 1986), where plain verbs do not inflect for person or number. Agreement verbs can modify movement, hand orientation, or both so that the verb is directed from the subject locus to the object locus. These loci may pick up physically present referents or abstract discourse points; TİD also licenses null arguments, whose distribution interacts with verb type (Sevinç, 2006).

The rewriting module assigns each argument a locus in signing space (1 the signer, 2 the addressee, 3a, 3b, . . . for third-person referents, introduced by a pointing sign IX after the referring noun) and annotates each verb from a 60-verb agreeing-verb inventory compiled from Kubuş (2008) (Appendix A) with its subject and object loci: ADAM KADIN GÖRMEK ('The man saw the woman') becomes ADAM IX_{3a} KADIN IX_{3b} $_{3a}\text{GÖRMEK}_{3b}$, and SEN BEN GÖRMEK ('You saw me') becomes $_{2}\text{GÖRMEK}_{1}$ without pointing signs (Examples (1)–(2)).

The rewriting operates on the gloss string before dictionary lookup. At synthesis time, the verb is first resolved like any other gloss, including sense disambiguation (Section 4.3); the selected clip is placed in a neutral canonical frame, its rest phases are trimmed, and the wrist trajectory and orientation are adjusted according to the verb's inventory category to direct the movement between loci. Pointing signs are realized as index-finger extensions toward the locus. The path is applied whichever sense is selected (for ARAMAK, both 'search' and 'call').

4.3 Context-Aware OOV Resolution

Because the TİD Sözlük inventory does not cover every gloss the LLMs produce (inflected or derived forms, compounds, proper nouns, lemmas outside the 2,000 entries) and some entries are polysemous, naive lookup drops tokens or returns an arbitrary sense. The resolver instead applies four ordered steps to every generated token, and every successful match re-enters Step 1 so that polysemous matches still pass through Sense Disambiguation (summary in Appendix D).

Step 1: Direct Lookup with Sense Disambiguation. The token is looked up verbatim in TİD Sözlük. If the entry is monosemous, its sign is returned directly. If the entry is polysemous, i.e., TİD Sözlük records more than one distinct sign for the same Turkish word, the resolver invokes **Sense Disambiguation**. For each candidate sign, we concatenate the dictionary description and example sentence, embed the result with *paraphrase-multilingual-mpnet-base-v2* (Reimers and Gurevych, 2019), and select the sense with highest cosine similarity to the full input Turkish sentence. The index is built from the dictionary only; each test sentence's own example text is excluded, so a test sentence is never matched against itself. This produces a per-occurrence sense decision conditioned on context rather than a corpus-wide default.

Step 2: Lexical Normalization. If Direct Lookup fails, the resolver applies deterministic rewrite rules: compound splitting on ^, joint-entry

comma-alias matching, conservative spelling reduction, inflectional suffix stripping, and semantic-prefix lookup. Each successful rewrite re-enters Step 1, so polysemous normalized forms still benefit from Sense Disambiguation.

Step 3: Semantic Nearest-Neighbor. When no normalized form succeeds, the token, with two words of context on either side when it occurs in the Turkish sentence and bare otherwise, is embedded with the same *paraphrase-multilingual-mpnet-base-v2* model and compared with precomputed embeddings of the bare dictionary gloss strings. The closest match is accepted only if cosine similarity ≥ 0.80. Accepted tokens typically share a root or near-synonym relation with the retrieved entry (e.g., PARLAMAK 'to gleam' resolves to PARLAK 'bright/shiny'). The result also re-enters Step 1.

Step 4: Fingerspelling. Tokens with no dictionary coverage and no sufficiently close semantic neighbor are spelled letter-by-letter using the İşaretçe alphabet (Section 3). Some proper nouns, including personal names and brand names, as well as acronyms, are routed to this step.

Qualitative examples of each strategy are given in Appendix F.

4.4 Stitching and Synthesis

We stitch the resolved sign clips into a continuous pose sequence. Every clip is canonicalized into a signer-relative frame, invalid pose frames are rejected, collapsed hand frames repaired and citation-form rest cropped; transition frames are inserted between consecutive clips in proportion to the dominant-hand wrist displacement at the boundary, inactive hands are filled with a neutral hand pose, and torso landmarks are interpolated over time. Each sentence begins with arms released in a neutral position. Compounds such as BILMEKˆDEĞIL (KNOWˆNOT) are expanded into parts, with the following DEĞIL treated as a reduced clitic whose location assimilates to the previous sign.

4.5 Avatar Rendering

To visualize the output, the pose sequence drives a 3D avatar from the Rocketbox library (Gonzalez-Franco et al., 2020) (Figure 2), with finger poses from WiLoR (Potamias et al., 2025) run on the dictionary clips, which gives anatomically plausible finger angles (more than 30 degrees past joint limits in 0.7% of test frames, against 36% for MediaPipe). Rule-based non-manual signals (head nods at clause boundaries, a head tilt for negation, blinks, gaze during fingerspelling, mouthing) are added, but no non-manual grammar, role shift or classifier constructions are modeled. The renderer is a visualization tool, not a contribution of this paper; it produces the signer-study stimuli (Section 6; Appendix G).



Figure 2: Avatar rendering of test sentence *O işe gelmedi* ('He/she did not come to work'), gloss O IŞ GELMEKˆDEĞIL. One frame from each sign.

5 Findings

We evaluate on the 371 test sentences in two settings: *gloss translation*, scoring LLM outputs against gold TİD glosses, and *end-to-end pose synthesis*, scoring stitched poses against ground-truth-gloss stitches and actual TİD Sözlük videos. We also ablate spatial agreement and context-aware resolution. Section 6 adds an evaluation of the rendered videos by TİD signers.

5.1 Evaluation Metrics

Gloss translation. For RQ1, we score predicted glosses against TİD Sözlük references using **BLEU-4** (Papineni et al., 2002) and **chrF** (Popović, 2015), both corpus-level sacreBLEU scores (Post, 2018) with tokenization disabled, as recommended for gloss output (Müller et al., 2023); **ROUGE** (Lin, 2004) at ROUGE-1, ROUGE-2, and ROUGE-L on gloss tokens; and **semantic similarity**, the mean cosine similarity between *paraphrase-multilingual-mpnet-base-v2* embeddings of predicted and reference glosses (Reimers and Gurevych, 2019). Systems are compared with a paired bootstrap over sentences (2,000 resamples) on corpus BLEU-4.

Pose synthesis. For RQ2, we compare pipeline poses with a target sequence using **MPJPE** (mean per joint position error) and **DTW** distance (Berndt and Clifford, 1994) over **All**, **Hands** and **Pose** (body) landmarks; lower is better. Sequences are aligned for scale and translation using the shoulders; MPJPE resamples both sequences to a com-

mon length, and DTW is normalized by warping-path length (Appendix E). These are proxy measures of pose alignment, not of comprehensibility.

5.2 Gloss Translation

Table 1 reports Turkish-to-TİD gloss translation across all ten LLM configurations; the dashed midrule separates fine-tuned open LLMs from prompted closed LLMs.

LLM glosses differ from the references in form, less in content. Even the best systems reach only about 13 BLEU-4 against the TİD Sözlük glosses, while semantic similarity is high for every system (0.887 to 0.913): the content is usually right, but lemma choice, ordering and repetition differ from the reference, which is itself sometimes a free translation (Appendix H). All systems produce shorter gloss sequences than the references (brevity penalty 0.75 to 0.88), the fine-tuned models slightly more so (0.75 to 0.83) than the closed models (0.78 to 0.88).

The larger fine-tuned open LLMs beat prompted closed LLMs. Gemma 3 12B (13.30 BLEU-4), Gemma 3 27B (13.29) and Qwen 3 32B (12.38) outperform the strongest prompted closed model, Gemini 3 Flash ten-shot (10.11), and GPT-5.4 ten-shot (9.37) on BLEU-4, chrF and every ROUGE column; a paired bootstrap confirms each of the six BLEU-4 differences (2.3 to 3.9 points, $p \leq .002$). The three large fine-tuned models are on par with each other: 12B and 27B tie on BLEU-4 and chrF, Qwen 3 32B leads on chrF, and 27B leads on ROUGE and semantic similarity. The advantage does not hold across the board: the 4B and 8B models beat GPT-5.4 ten-shot ($p \leq .021$) but not Gemini 3 Flash ten-shot ($p \geq .09$), and on semantic similarity the ten-shot closed models (0.903 and 0.909) are level with the fine-tuned ones (0.887 to 0.913). The closed models' sizes are not disclosed, so we do not interpret the comparison in terms of size. We attribute the advantage of the larger fine-tuned models to TİD gloss conventions (ordering, citation lemmas, compound markers, omitted inflection), which fine-tuning on 2,827 pairs learns more reliably than ten random in-context examples, consistent with Zhang et al. (2023); retrieved or dictionary-augmented examples (Tanzer et al., 2024b) might close part of the gap.

Within Gemma 3, BLEU-4 rises from 4B to 12B and then levels off; ten-shot prompting improves both closed models over one-shot (by 1.55 and 1.28 BLEU-4), but the gain is small relative to the gap to the larger fine-tuned models.

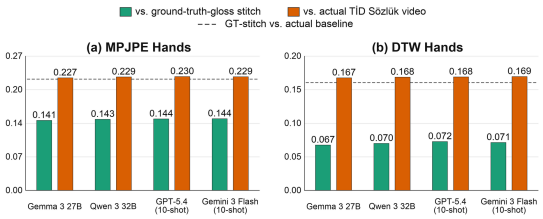


Figure 3: Pose metrics (lower is better). Teal: predicted-gloss stitch vs. ground-truth-gloss stitch. Orange: full pipeline vs. actual TİD Sözlük video. Dashed line: ground-truth-gloss stitch vs. actual video.

5.3 End-to-End Pose Synthesis

We run the full pipeline with Gemma 3 27B, Qwen 3 32B, and the 10-shot GPT-5.4 and Gemini 3 Flash. We compare predicted-gloss poses with ground-truth-gloss poses passed through the same stitching pipeline, which isolates gloss prediction errors, and with the actual TİD Sözlük sentence videos, which also includes the gap between dictionary stitching and natural signing (Figure 3; full results in Appendix E).

Gloss prediction adds little pose error on top of synthesis. Predicted-gloss stitches differ from ground-truth-gloss stitches (teal bars; MPJPE Hands 0.141 to 0.144, about 63% of the distance between the ground-truth stitch and the real video), so the difference is not negligible in itself. Against the real video (orange bars), however, the predicted-gloss stitches are only slightly farther than the ground-truth-gloss stitch (MPJPE Hands 0.227 to 0.230 vs. 0.224; DTW Hands 0.167 to 0.169 vs. 0.160). Gemma 3 27B is best on all six metrics among the four systems run through the pose pipeline, and system differences are small (MPJPE-All within 0.0036). Two readings are possible: gloss errors add little to the distance from real signing compared with dictionary stitching itself, or the metrics saturate once every stitched sequence is far from the reference, so that a real difference between gloss sources is compressed.

A substantial gap remains between dictionary stitching and natural signing. Against actual videos (orange bars), Gemma 3 27B again leads among model outputs, but even the ground-truth gloss stitch (DTW Hands ≈ 0.16, MPJPE Hands ≈ 0.22) is far from zero. This gap reflects timing,

Model	Setting	BLEU-4	chrF	R-1	R-2	R-L	Sem.
Gemma 3 4B	fine-tuned	$11.08_{\pm 1.4}$	$43.4_{\pm 1.6}$	0.526	0.216	0.471	0.895
Gemma 3 12B	fine-tuned	$\mathbf{13.30}_{\pm 1.4}$	$46.6_{\pm 1.3}$	0.578	0.249	0.515	0.909
Gemma 3 27B	fine-tuned	$13.29_{\pm 1.3}$	$46.5_{\pm 1.3}$	**0.587**	**0.253**	**0.527**	**0.913**
Qwen 3 4B	fine-tuned	$11.23_{\pm 1.3}$	$43.1_{\pm 1.3}$	0.533	0.220	0.478	0.892
Qwen 3 8B	fine-tuned	$10.96_{\pm 1.3}$	$43.1_{\pm 1.4}$	0.534	0.218	0.478	0.887
Qwen 3 32B	fine-tuned	$12.38_{\pm 1.3}$	$\mathbf{46.8}_{\pm 1.2}$	0.563	0.229	0.507	0.910
Gemini 3 Flash	1-shot	$8.56_{\pm 1.1}$	$43.9_{\pm 1.2}$	0.551	0.186	0.479	0.902
Gemini 3 Flash	10-shot	$10.11_{\pm 1.2}$	$45.2_{\pm 1.2}$	0.558	0.195	0.490	0.903
GPT-5.4	1-shot	$8.09_{\pm 1.2}$	$44.3_{\pm 1.0}$	0.539	0.173	0.469	0.904
GPT-5.4	10-shot	$9.37_{\pm 1.2}$	$45.5_{\pm 1.1}$	0.552	0.186	0.484	0.909

Table 1: Turkish-to-TİD gloss translation on the 371-sentence test set: fine-tuned open LLMs (top) and prompted closed LLMs (bottom). BLEU-4 and chrF are corpus-level sacreBLEU scores with tokenization disabled (Müller et al., 2023); subscripts are 95% bootstrap CI half-widths. Higher is better; best per column in bold.

coarticulation, prosody, and signer-specific realization absent from isolated lexical clips.

5.4 Ablations: Agreement and Context-Aware Resolution

For RQ2, we disable each module in turn and compare **DTW Hands** against the actual video (paired Wilcoxon tests, Holm-corrected across systems), each on the subset of sentences where the module can change the output. The second ablation, *context-aware resolution*, disables Sense Disambiguation, Semantic Nearest-Neighbor and Fingerspelling together, so it also covers in-vocabulary polysemous tokens; the exact configuration and fallbacks are given in Appendix E.

Spatial agreement gives a small, significant improvement in hand-trajectory alignment. On the 169 test sentences whose reference gloss contains an agreeing verb, disabling agreement worsens DTW Hands by 0.0003 to 0.0018, about 1% of the DTW Hands value of 0.167. The effect is significant for Gemma 3 27B ($p_{\text{Holm}} = .0063$), GPT-5.4 (.0060), Gemini 3 Flash (.0003) and the ground-truth gloss stitch ($\Delta = 0.0019$, $p_{\text{Holm}} < .0001$), but not for Qwen 3 32B (.11). The effect is reliable but small, and signers still flagged agreement-verb paths (Section 6), so we do not claim that it is visible to viewers.

Context-aware resolution is the larger contributor. OOV handling (Steps 2 to 4) applies to 10.6% of Gemma 3 27B tokens and 10.2% of Qwen 3 32B tokens, and another 7.5% of tokens are polysemous and pass through Sense Disambiguation (Appendix D); among OOV tokens, Semantic Nearest-Neighbor accounts for about 44%, Lexical Normalization 31% and Fingerspelling 26%. On sentences where Sense Disambiguation or Semantic Nearest-Neighbor was used (261 to 281 per system), disabling the three context-aware steps worsens DTW Hands by 0.0038 to 0.0047 for all four systems ($p_{\text{Holm}} \leq 3.5\times10^{-7}$) and by 0.0051 for the ground-truth gloss stitch ($p_{\text{Holm}} < .0001$), about 2.5% of the DTW Hands value; the result is unchanged when sentences affected by test examples in the sense index are removed (Appendix D). Part of the effect comes from Sense Disambiguation on in-vocabulary tokens, and the two modules are measured on different sentence subsets. MPJPE Hands is not significant for either module, which we attribute to temporal misalignment under linear resampling (Appendix E). Context-aware resolution improves coverage but cannot repair upstream glossing errors (Appendix F).

(the result is unchanged when sentences affected by test examples in the sense index are removed; Appendix D)

Results hold with a second hand estimator. All pose metrics use MediaPipe hands. Rescoring with WiLoR reference hands (Potamias et al., 2025) lowers the Hands errors of every system by 3 to 6% and leaves the system ranking and the direction and significance of both ablation effects unchanged, but the effect sizes change: the agreement effect halves for Gemma 3 27B and the context-aware resolution effect is 13 to 25% smaller (Appendix I).

6 Evaluation with TİD Signers

Automatic metrics show where generated poses differ from real signing, but not whether signers understand the result. We therefore ran a diagnostic study with Deaf and expert TİD signers to find which stage of the pipeline limits the output.

Design. We sampled 25 of the 371 test sentences at random (median 13 Turkish words, range 3 to

37; 22 are multi-clause and 13 consist of two or more sentences). Each sentence was shown as **our avatar** (glosses from fine-tuned Gemma 3 27B), a **gold-gloss avatar** (reference glosses, same synthesizer), and the **real signer** (TİD Sözlük video); the first two differ only in glossing, the last two in everything after glossing. Deaf signers saw each sentence once, as our avatar or as the real signer (two counterbalanced lists). Four expert signers (one Deaf, three hearing, researchers or professional translators) rated all three versions. We report understanding, hand signs and naturalness on 1 to 5 scales. Before seeing the Turkish sentence, experts wrote a free translation of the first video of each item. Appendix H gives the full procedure.

The real signer is far above both avatars. Both groups rated the real signer near the top of the scale and our avatar much lower (Table 2a); Deaf raters understood at least half of the sentence for 31% of avatar videos. Experts most often tagged the avatar as looking artificial (78%) and as missing non-manual signals (63%), and named errors in handshape, location and orientation, and citation forms where TİD would use a classifier construction. Seeing the real signer in the same item and Deaf raters' lower acceptance of synthetic avatars may lower the absolute ratings (Kacorri et al., 2013; Quandt et al., 2022), but both apply equally to our avatar and the gold-gloss avatar.

LLM glossing has little effect on the ratings. Experts rated our avatar 2.40 and the gold-gloss avatar 2.52 (Table 2b; paired difference over the 79 rated items -0.09, 95% CI -0.25 to 0.06, Wilcoxon $p = .25$), with similar TİD grammar ratings (2.25 and 2.24). The difference is small compared with the 2.5-point gap to the real signer, in line with the pose metrics (Section 5.3).

Experts recovered part of the meaning in blind translation. We scored the experts' 55 free translations (one per item for the first video shown) against the source sentence with the semantic similarity model of Table 1 and with chrF, using the other 24 sources as a chance baseline. The translations reached a similarity of 0.72 (chance 0.18) and a chrF of 34.6 (chance 10.3), and were closest to their own source in 50 of 55 cases (Table 2c); the 33 translations made from our avatar reached 0.77 (chrF 41.1). Some content was also recovered from videos that the same expert rated 1 or 2 on understanding (35 videos, similarity 0.59). Glossing is therefore not the main bottleneck; sign synthesis, the use of signing space and non-manual grammar need the most work.

(a) Real signer vs. our avatar

	Real	Avatar
Deaf signers ($n = 4$)	4.82	1.86
Experts ($n = 4$)	4.96	2.40

(b) LLM vs. gold glosses (experts, 79 pairs)

	Gold	LLM
Rating (mean of three)	2.52	2.40
TİD grammar	2.24	2.25

(c) Blind translation (experts, 55 translations)

	Own source	Chance
Semantic similarity	0.72	0.18
chrF	34.6	10.3

Table 2: Signer study, 25 sentences. Ratings are the mean of understanding, hand signs and naturalness (1 to 5), averaged per rater. Per-scale ratings, ranges and problem tags are in Appendix H.

7 Discussion and Conclusion

We presented, to our knowledge, the first modular Turkish-text-to-TİD pipeline combining LLM-based glossing, TİD-specific pose synthesis, and avatar rendering. The larger fine-tuned open models (Gemma 3 12B and 27B, Qwen 3 32B) beat prompted closed models on BLEU, chrF and ROUGE, though not on semantic similarity; we use Gemma 3 27B, the best on semantic similarity and on the pose metrics among the systems run through the pose pipeline, downstream. Both linguistic modules give small but significant improvements in hand-trajectory alignment, context-aware resolution for every system and agreement for three of four systems and the gold stitch; signers still noted errors in agreement-verb paths, so this gain is not yet visible to them.

Pose metrics and human ratings locate the bottleneck in the same place: LLM glosses perform close to gold glosses, but both are far from real signing, mainly because of non-manual signals, sign form and the use of space. This agrees with ASL generation work that stresses non-manual markers (Zhang et al., 2025). The modular design made this diagnosis possible and can be applied to other under-resourced sign languages with a dictionary corpus: dictionary resources, linguistic rules, and adaptable LLMs are practical for glossing, while natural sign synthesis and non-manual grammar remain the main open problems.

8 Limitations

This study has several limitations. First, our human evaluation is a small diagnostic study with four Deaf and four expert raters on 25 sentences. The system does not yet model non-manual grammar such as brow movement in questions and conditionals, and constructions such as role shift, classifiers or body lean, and experts noted missing non-manual signals in most avatar videos.

Second, the pose-stitching pipeline depends on TİD Sözlük and related lexical resources. Signs come from citation-form dictionary clips, so coarticulation, sign variants and directional movement remain approximate. This makes the system practical in a low-resource setting, but constrains the output to dictionary-style lexical signs. The actual-video comparison shows that even ground-truth gloss stitches differ from natural sentence videos, reflecting missing coarticulation, prosody, non-manual markers, discourse structure, and signer-specific timing.

Third, the spatial agreement and OOV modules are deterministic approximations. They do not model the full linguistic range of TİD agreement, classifier constructions, role shift, or context-dependent reference tracking, and OOV substitution may be imperfect in specific discourse contexts. Future work could also add an iconicity-aware OOV strategy. Rather than resolving every missing lexical item through textual nearest-neighbor substitution or fingerspelling, the resolver could compose provisional visually motivated forms when appropriate, using iconic mappings between meaning and articulatory form, such as arm movement for TO-FLY or a claw/round handshape for classifying an object like BALL (Keleş et al., 2026).

Fourth, the ablations measure each module against the full pipeline on the subset of sentences it affects; we did not run a configuration with both modules disabled, so the gain over fully naive dictionary stitching is not measured directly, and the context-aware resolution ablation also removes sense disambiguation of in-vocabulary tokens.

Finally, the model comparison is limited to the available 371-sentence test set and selected LLM configurations. The prompted closed models received one or ten random in-context examples; retrieval-augmented or dictionary-augmented prompting (Tanzer et al., 2024b) is a stronger baseline that we have not run, so the advantage of fine-tuning may be smaller against it. Test and training sentences come from the same dictionary and signers, so results may change with out-of-domain text, larger corpora, better sign synthesis, or evaluation sets targeting phenomena such as classifier predicates, negation, fingerspelling, and regional variation.

Ethics Statement

The signer study was approved by the ethics board of the first author's university (details withheld for review). All participants gave informed consent; they were recruited through the Deaf community and through colleagues working with TİD, took part online in a single short session, and could stop at any time.

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A TİD Agreement Verb Inventory

Table 3 lists the verbs in the TİD agreement inventory. Examples (1) and (2) illustrate the rewriting of Section 4.2 for third-person referents and for first/second-person pronouns.

(1) a. ADAM KADIN GÖRMEK
MAN WOMAN SEE
'The man saw the woman.'

b. ADAM IX_{3a} KADIN IX_{3b} $_{3a}$GÖRMEK$_{3b}$
MAN IX_{3a} WOMAN IX_{3b} $_{3a}$SEE$_{3b}$
'The man saw the woman.'

(2) a. SEN BEN GÖRMEK
YOU I SEE
'You saw me.'

b. $_{2}$GÖRMEK$_{1}$
$_{2}$SEE$_{1}$
'You saw me.'

Category 1: movement and orientation (40 verbs)					
ALAY ETMEK	*mock*	ALMAK	*take*	AVANTAJ SAĞLAMAK	*benefit (s.o.)*
AYRILMAK	*separate*	BAĞIRMAK	*shout at*	BAKMAK	*look at*
BAŞVURMAK	*apply to*	BIÇAKLAMAK	*stab*	ÇAĞIRMAK	*call/summon*
DESTEKLEMEK	*support*	DÖVMEK	*beat*	DURDURMAK	*stop*
E-POSTA ATMAK	*email*	EMRETMEK	*order*	ETKILEMEK	*affect*
FAKS GÖNDERMEK	*fax*	GÖNDERMEK	*send*	GÖRMEK	*see*
IĞNELEMEK	*tease/sting*	ISIRMAK	*bite*	KARŞI OLMAK	*oppose*
KOPMAK	*break off*	KOPYA ÇEKMEK	*copy/cheat*	KOVMAK	*dismiss*
KÖTÜLEMEK	*slander*	KÜFÜR ETMEK	*curse at*	ÖDEMEK	*pay*
ÖĞRETMEK	*teach*	ÖLDÜRMEK	*kill*	ÖPMEK	*kiss*
RICA ETMEK	*request*	SEÇMEK	*choose*	SESLENMEK	*call out to*
SUSTURMAK	*silence*	ŞIKAYET ETMEK	*complain*	TAKIP ETMEK	*follow*
TEKLIF ETMEK	*offer*	VURMAK	*hit*	YENMEK	*defeat*
ZORLAMAK	*force*				
Category 2: orientation only (3 verbs)					
HOŞLANMAK	*like*	PAYLAŞMAK	*share*	SORGULAMAK	*question*
Category 3: trajectory only (17 verbs)					
ANLATMAK	*explain*	ARAMAK	*call/search*	BESLEMEK	*feed*
BORÇ VERMEK	*lend*	CEVAPLAMAK	*answer*	DAVET ETMEK	*invite*
HABER VERMEK	*notify*	KUTLAMAK	*congratulate*	ÖDÜL VERMEK	*award*
SATMAK	*sell*	SÖYLEMEK	*tell*	SORU SORMAK	*ask*
SUÇ ATMAK	*accuse*	TEŞEKKÜR ETMEK	*thank*	VERMEK	*give*
YÖNETMEK	*manage*	ZULÜM ETMEK	*oppress*		

Table 3: TİD agreement verb inventory (60 verbs) taken from (Kubuş, 2008). Category 1 modifies both wrist trajectory and wrist-yaw orientation; Category 2 modifies orientation only; Category 3 modifies trajectory only.

B Full System Message

```
You are a model that converts Turkish
↪  sentences into Turkish Sign Language
↪  (TİD) gloss sequences.

Task:
Given one Turkish sentence, output
↪  exactly one line containing the
↪  corresponding TİD glosses.

Output constraints:
- Output only the gloss sequence.
- Separate glosses with single spaces.
- Do not add explanations, punctuation,
↪  quotes, or extra lines.

Gloss conventions:
- Use ALL-CAPS Turkish lemma/root
↪  glosses.
- Do not output lowercase forms or
↪  Turkish inflectional suffixes.
- Avoid Turkish participial or suffixed
↪  surface forms as one-off gloss
↪  tokens when the line should use
↪  standard citation lemmas instead.
- Prefer TİD-style information
↪  structure:
  - time/topic first when present
  - then Subject-Object-Verb when the
  ↪  clause has agent and patient
- Omit tense, agreement, person, and
↪  case morphology when they are not
↪  lexical signs.
- Use `^` only for fixed lexical
↪  compounds or lexicalized negation.
↪  For lexicalized negation the usual
↪  form is one compound token: the
↪  negated lemma, then `^`, then DEĞİL
↪  (i.e. LEMMA^DEĞİL).
- Do not output optional alternatives or
↪  parenthesized material.
- If a source expression is idiomatic,
↪  output the most natural TİD gloss
↪  sequence rather than a word-for-word
↪  mapping.
```

C Fine-tuning Hyperparameters

Table 4 lists the hyperparameters shared across all fine-tuned open models. Target modules are the attention and feed-forward projection matrices.

Hyperparameter	Value
Precision	bf16
LoRA rank	64
LoRA α	16
LoRA dropout	0.1
Learning rate	2×10^{-4}
Optimizer	AdamW
Warmup ratio	0.03
Training epochs	3
Max sequence length	1024

Table 4: Fine-tuning hyperparameters shared across all open models.

Data, training and decoding. The 371 test sentences are fixed and shared by all systems. All other TİD Sözlük Turkish-gloss example pairs whose gloss has at least two tokens are split at random into training (85%) and validation (15%) sets (seed 42), giving 2,827 training and 500 validation pairs. Each training example consists of the system message (Appendix B), the Turkish sentence, and the gold gloss. Training uses gradient accumulation over 4 steps, weight decay 0.001, gradient clipping at 0.3, and evaluation on the validation set every 50 steps; the checkpoint with the lowest validation loss is kept. The fine-tuned models used greedy decoding. The closed models are queried with temperature 0 (honored by both APIs for these runs); GPT-5.4 is run with reasoning effort set to `low`, and Gemini 3 Flash with its default thinking level. The one and ten in-context examples are selected at random from the training split.

D OOV Handling Statistics

Tables 5 and 6 report OOV resolution statistics. Table 5 shows mean strategy counts across Gemma 3 27B and Qwen 3 32B; Table 6 shows per-model OOV rates for all four systems. Rows above the first rule in Table 5 are in-vocabulary resolutions (monosemous tokens found verbatim, and polysemous tokens found verbatim and resolved by Sense Disambiguation); rows below require OOV handling. The five rows are disjoint and sum to the total.

Resolution strategy	Mean Count	% of OOV	% of Total
Direct Lookup (monosemous)	4128	–	82.1%
Direct Lookup + Sense Disambiguation	377	–	7.5%
Lexical Normalisation	160	30.6%	3.2%
Semantic Nearest-Neighbour	228	43.6%	4.5%
Fingerspelling	135	25.8%	2.7%
Total Generated Tokens	5028	–	100.0%

Table 5: Mean resolution strategy distribution across Gemma 3 27B and Qwen 3 32B (≈5,028 tokens per system). The two Direct Lookup rows count tokens found verbatim in TİD Sözlük, split into monosemous entries and polysemous entries resolved by Sense Disambiguation; the rows are disjoint ($4128 + 377 + 160 + 228 + 135 = 5028$).

Model	Total Tokens	OOV Count	% OOV
Gemma 3 27B	4,965	525	10.6%
Qwen 3 32B	5,091	519	10.2%
GPT-5.4	5,536	666	12.0%
Gemini 3 Flash	5,190	562	10.8%

Table 6: Per-model OOV rates across the 371 test sentences for all four pipeline systems.

Sense index and the test set. The pose results in Section 5.4 were produced with a sense index that still contained the example sentences of the test items, so for the 37 test sentences that are the example of a polysemous entry (23 to 24 per system in which the headword is predicted), the query was compared against text containing itself. Rebuilding the index without the test examples changes 5 to 6% of sense decisions (21 of 380 for Gemma 3 27B, 19 of 407 for Qwen 3 32B, 27 of 447 for the gold gloss), in 15 to 22 sentences per system. Removing these sentences from the ablation subset leaves the effect of disabling context-aware resolution unchanged (Gemma 3 27B 0.0047 to 0.0048, Qwen 3 32B 0.0041 to 0.0042, GPT-5.4 0.0038 to 0.0038, Gemini 3 Flash 0.0042 to 0.0041, gold stitch 0.0051 to 0.0052; all $p < 10^{-6}$).

E Detailed Pose Evaluation Results

Metric definitions. All metrics are computed on 2D (x, y) landmark coordinates. Before scoring, the predicted sequence is scaled and translated so that its mean shoulder width and shoulder midpoint match those of the reference sequence; reference hand landmarks that the pose estimator did not detect are excluded from both metrics (frames in which a reference hand is missing are filled with the hip position and masked out). Since the stitched and the real sequences differ in length, MPJPE is computed after linearly resampling both sequences in time to the longer of the two lengths, T:

$$\text{MPJPE} = \frac{1}{|V|} \sum_{(t,j)\in V} \|p_j^t - \hat{p}_j^t\|_2, \tag{1}$$

where p_j^t and $\hat{p}_j^t$ are the reference and predicted positions of joint j at resampled frame t and V is the set of (frame, joint) pairs with a valid reference landmark. Because the resampling is linear, MPJPE penalizes any shift in timing between the two sequences. DTW (Berndt and Clifford, 1994) instead aligns the sequences in time:

$$\text{DTW}(X, Y) = \frac{1}{|\pi^*|} \sum_{(i,j)\in\pi^*} d(x_i, y_j), \quad d(x_i, y_j) = \frac{1}{|V_i|} \sum_{k\in V_i} \|x_{i,k} - y_{j,k}\|_2, \tag{2}$$

where X and Y are the reference and predicted pose sequences, π^* is the minimum-cost warping path, d is the mean Euclidean distance over the valid joints V_i of reference frame i, and the sum is divided by the path length $|\pi^*|$ so that the value does not grow with sequence length. Both metrics are therefore in units of normalized image coordinates per joint.

Ablation configurations. The agreement ablation turns off locus insertion and verb-path rewriting (Section 4.2) and is scored on the 169 test sentences whose reference gloss contains a verb from the agreement inventory. The context-aware resolution ablation turns off Sense Disambiguation (Step 1), Semantic Nearest-Neighbor (Step 3) and Fingerspelling (Step 4) of the resolver, while Direct Lookup and Lexical Normalization (Step 2) stay on; it is scored on the sentences where Sense Disambiguation or Semantic Nearest-Neighbor fired in the full pipeline (261 to 281 per system). With Sense Disambiguation off, a polysemous token falls back to the longest of its dictionary clips; a token that only Steps 3 or 4 could resolve is dropped from the sequence. We did not run a configuration with both modules off, so the two effects are reported separately and not as a gain over fully naive stitching.

Why MPJPE does not reflect the ablations. MPJPE compares frame t of one sequence with frame t of the other after linear resampling to a common length. A dropped token, an inserted pointing sign or a changed verb path shifts the timing of everything that follows it, and this temporal misalignment adds framewise error that masks the local improvement in the hand path. DTW aligns the sequences in time before measuring distance, so it is the metric that can register these changes; this is why the ablation effects are visible in DTW Hands but not in MPJPE Hands.

This appendix also reports the full per-landmark MPJPE and DTW behind Figure 3. Table 7 compares pose sequences generated from each system's predicted glosses against pose sequences generated from the ground-truth glosses, with both sides passed through the same pose-stitching pipeline. Table 8 compares the full pipeline outputs against actual TİD Sözlük sentence videos. MPJPE and DTW are reported for all skeletal points (All), hands (H), and body landmarks (P). For all metrics, lower is better; the best value per column is in bold.

Model	MPJPE-All	MPJPE-H	MPJPE-P	DTW-All	DTW-H	DTW-P
Gemma 3 27B	**0.1125**	**0.1412**	**0.0644**	**0.0564**	**0.0673**	**0.0369**
Qwen 3 32B	0.1146	0.1433	0.0663	0.0586	0.0696	0.0391
GPT-5.4	0.1159	0.1440	0.0686	0.0616	0.0724	0.0425
Gemini 3 Flash	0.1161	0.1445	0.0683	0.0603	0.0710	0.0415

Table 7: Pose evaluation against the *ground-truth-gloss stitch*, i.e., the pose sequence obtained by stitching the reference TİD Sözlük glosses through the same pipeline. This comparison isolates the effect of gloss prediction errors under a shared synthesis pipeline.

Model	MPJPE-All	MPJPE-H	MPJPE-P	DTW-All	DTW-H	DTW-P
GT gloss stitch	**0.1799**	**0.2245**	**0.1049**	**0.1344**	**0.1599**	**0.0873**
Gemma 3 27B	0.1817	0.2269	0.1056	0.1405	0.1669	0.0900
Qwen 3 32B	0.1833	0.2287	0.1071	0.1409	0.1679	0.0908
GPT-5.4	0.1848	0.2297	0.1094	0.1422	0.1684	0.0932
Gemini 3 Flash	0.1843	0.2293	0.1087	0.1422	0.1689	0.0925

Table 8: Pose evaluation against *actual TİD Sözlük sentence videos*. The first row reports the ground-truth-gloss stitch's own distance to the actual video, i.e., the irreducible floor for any dictionary-style stitching pipeline; subsequent rows report each LLM-driven pipeline.

F Model Output Examples and Error Analysis

Table 9 reports four test-set sentences in which the glosses predicted by Gemma 3 27B require non-trivial resolution before synthesis: two polysemy cases handled by Sense Disambiguation (in-vocabulary tokens) and two OOV cases. The purpose of these examples is not to evaluate Gemma's overall translation quality, but to show how the downstream resolver handles predicted tokens that are missing from the TİD Sözlük inventory. Each predicted token was passed through the same resolver used by the synthesizer (Section 4.3); the annotations in the rightmost column report the strategy selected for the relevant OOV token and the dictionary entry to which it was resolved.

Strategies in Examples. In Ex. 1 and Ex. 2 Gemma 3 27B predicts ARAMAK (to call) in both cases but the context differs; Sense Disambiguation selects a different sign variant for each (search vs. telephone call). Ex. 3 illustrates Semantic Nearest-Neighbour: PARLAMAK (to gleam) is absent from TİD Sözlük but is resolved to the adjective PARLAK (bright/shiny), which shares the same lexical root (cosine similarity 0.95). Ex. 4 shows Fingerspelling applied to the city name MARDIN, which has no dictionary sign.

Ex. 1	TR: *Ceket ve gömlek aldıktan sonra çizgili kravat aradım. Biraz aradıktan sonra bulunca iki tane aldım.* EN: *After buying a jacket and shirt I looked for a striped tie. After searching a while I found one and bought two.*
Reference	CEKET GÖMLEK BEN ALMAK KRAVAT ÇİZGİLİ BAKMAK ARAMAK ARAMAK BULMAK İKİ TANE ALMAK GL: JACKET SHIRT 1SG BUY TIE STRIPED LOOK SEARCH SEARCH FIND TWO BUY
Gemma 3 27B	CEKET GÖMLEK ALMAK SONRA KRAVAT ÇİZGİLİ ARAMAK ARAMAK BULMAK İKİ TANE ALMAK GL: JACKET SHIRT BUY THEN TIE STRIPED SEARCH SEARCH FIND TWO BUY
Sense	ARAMAK is polysemous (4 senses in TİD Sözlük); Sense Disambiguation selects the *search for something* variant.
Ex. 2	TR: *Sokakta yürürken yaşlı bir adamın düştüğünü gördüm. Hemen ambulansı aradım ve beş dakikada geldiler.* EN: *While walking on the street I saw an elderly man fall. I immediately called an ambulance and they arrived in five minutes.*
Reference	BEN SOKAK YÜRÜMEK YÜRÜMEK YAŞLI YÜRÜMEK DÜŞMEK BEN HEMEN AMBULANS ARAMAK BEŞ DAKİKA GELMEK GL: 1SG STREET WALK WALK ELDERLY WALK FALL 1SG IMMEDIATELY AMBULANCE CALL FIVE MINUTE ARRIVE
Gemma 3 27B	BEN SOKAK BEN YÜRÜMEK YÜRÜMEK YAŞLI ADAM DÜŞMEK BEN AMBULANS ARAMAK BEŞ DAKİKA GELMEK GL: 1SG STREET 1SG WALK WALK ELDERLY MAN FALL 1SG AMBULANCE CALL FIVE MINUTE ARRIVE
Sense	ARAMAK is polysemous; Sense Disambiguation selects the *call someone by phone* variant.
Ex. 3	TR: *Evimin duvarları çok kirlendiği için yeni boya aldım ve boyadım. Şimdi çok temiz oldu, her yer parlıyor.* EN: *My house walls had gotten very dirty so I bought new paint and painted them. Now it is very clean and everything is gleaming.*
Reference	BEN EV DUVAR HEPSİ KİRLENMEK BEN EN İYİ YENİ BOYA ALMAK BOYAMAK BOYAMAK TEMİZ PARLAMAK GL: 1SG HOUSE WALL ALL DIRTY 1SG BEST NEW PAINT BUY PAINT PAINT CLEAN GLEAM
Gemma 3 27B	BEN EV DUVAR KİR BEN BOYA ALMAK YENİ BOYA TEMİZLEMEK TEMİZLEMEK PARLAMAK GL: 1SG HOUSE WALL DIRT 1SG PAINT BUY NEW PAINT CLEAN CLEAN GLEAM
OOV	PARLAMAK → Semantic Nearest-Neighbour → PARLAK (cosine sim. 0.95; verb form resolved to dictionary adjective of same root).
After OOV	BEN EV DUVAR KİR BEN BOYA ALMAK YENİ BOYA TEMİZLEMEK TEMİZLEMEK PARLAK GL: 1SG HOUSE WALL DIRT 1SG PAINT BUY NEW PAINT CLEAN CLEAN BRIGHT
Ex. 4	TR: *Otuz yıldır Mardin'de oturuyorum.* EN: *I have been living in Mardin for thirty years.*
Reference	BEN MARDİN OTUZ SENE OTURMAK GL: 1SG MARDIN THIRTY YEAR LIVE
Gemma 3 27B	BEN MARDİN YER OTUZ SENE OTURMAK GL: 1SG MARDIN PLACE THIRTY YEAR LIVE
OOV	MARDIN → Fingerspelling (city name; no dictionary sign).
After OOV	BEN [FS:MARDİN] YER OTUZ SENE OTURMAK GL: 1SG [M-A-R-D-I-N] PLACE THIRTY YEAR LIVE

Table 9: OOV resolution examples from the test set (Gemma 3 27B predictions). *Sense Disambiguation* selects the contextually appropriate sign variant for polysemous entries (Ex. 1 and 2: same gloss token ARAMAK, different sign video). *Semantic Nearest-Neighbour* resolves absent tokens to the nearest dictionary entry by embedding similarity (Ex. 3: PARLAMAK→PARLAK). *Fingerspelling* uses the İşaretçe alphabet for proper nouns and final-fallback items (Ex. 4). GL lines provide English gloss translations. Underlined tokens indicate polysemy resolution (rows labelled Sense) or OOV resolution (rows labelled OOV).

G Avatar Rendering Details

We use the Rocketbox avatar Business_Male_06 (MIT licence; Gonzalez-Franco et al., 2020), converted to glTF with its 52 facial shape keys. Following common guidelines for sign language video, the avatar wears a solid dark shirt in front of a plain light grey background. The study stimuli use glosses predicted by fine-tuned Gemma 3 27B. A browser page poses the avatar's skeleton frame by frame from the synthesized pose sequence; videos are rendered in headless Chromium (Playwright) and encoded with ffmpeg (libx264; 1440×1080, 24 fps, front view).

Non-manual signals. The avatar nods at clause boundaries, tilts the head backward for negation, blinks in pauses, and looks at the camera during fingerspelling. Mouthing is rule-based from the gloss text: the first syllable of nouns, adjectives, numbers and names is mouthed, and plain verbs are mouthed only when they are sentence-final or last at least 0.6 seconds. Pointing signs, pronouns, agreement verbs, DEĞIL, YOK, VE and AMA are not mouthed. About 68% of signs are mouthed. During fingerspelling the whole word is mouthed at half strength. Facial grammar (e.g., brow movement in questions and conditionals), role shift, classifier constructions and body lean beyond negation are not modelled.

H Signer Study Details

Participants. All participants took part online and gave informed consent; they were recruited through the Deaf community and through colleagues working with TİD. Table 10 lists the background of the Deaf participants. All four use TİD daily, and all attended a school for the Deaf. Table 11 lists the four experts.

ID	Age	TİD from	Deaf school
D1	30 to 39	birth	yes
D2	30 to 39	birth	yes
D3	30 to 39	birth	yes
D4	50 to 59	age 7 to 12	yes

Table 10: Deaf participants.

Procedure. Glosses for our avatar were predicted by fine-tuned Gemma 3 27B. Deaf participants saw each sentence once, either as our avatar (13 sentences) or as the real signer (12), in two counterbalanced lists; they did not see the gold-gloss avatar. They rated how much they understood each video (1 to 5) before the Turkish sentence was shown, then saw the sentence and rated meaning, correctness of the hand signs, and naturalness, and marked whether anything was missing or wrong (*none*, *some*, *much*). For each item, experts saw all three versions; before the Turkish sentence was shown they wrote a Turkish translation of what they understood from the first video. They then rated all versions, and tagged problems (e.g., missing non-manual signals, missing sign, word order). Raters translated and judged freely rather than choosing among options, which is more demanding than evaluations with isolated signs or multiple-choice items.

Analysis. Scores are averaged per rater; 95% confidence intervals come from a bootstrap over raters. Our avatar and the gold-gloss avatar are compared with a paired Wilcoxon test over expert ratings.

Results per expert. Table 11 gives the results per expert. Every expert rated the real signer far above both avatar versions (gap to our avatar 1.87 to 3.23 points). The difference between our avatar and the gold-gloss avatar is small for every expert (-0.32 to $+0.18$) and changes sign between experts. Among Deaf raters, the gap between the real signer and our avatar ranged from 2.54 to 3.79 points. Deaf raters marked *much* missing or wrong for 73% of avatar videos and for none of the real-signer videos, and rated their understanding at 3 or higher for all real-signer videos.

Blind translations. For each translation we computed semantic similarity to the source sentence, chrF and BLEU (sacreBLEU), and content-word recall (stopwords removed, words matched on their first four letters). As a chance baseline, each translation was scored against the other 24 source sentences. Over all 55 translations, BLEU is 15.0 and content-word recall 0.39 (chance 0.02; Wilcoxon $p < .001$); recall ranged from 0.27 to 0.54 between experts and was higher for short sentences (up to 13 words: 0.48) than for long ones (0.26). Semantic similarity and chrF both correlate with the understanding rating ($\rho = .81$, $n = 55$). Because each expert translated only one version, translation scores of our avatar and the gold-gloss avatar are confounded with the expert and are not compared.

Rating context. Two factors may lower the absolute avatar ratings: experts saw the real signer in the same item, which tends to lower ratings of signing animation (Kacorri et al., 2013), and Deaf signers tend to rate computer-synthesized avatars lower than hearing users (Quandt et al., 2022). Both apply equally to our avatar and the gold-gloss avatar. Deaf raters' understanding ratings of 3 or higher ranged from 0 to 77% per rater. Sentence length explains little of the gap: Deaf raters rated short sentences (up to 13 words) only slightly higher than long ones (1.96 vs. 1.77).

Sentence length and session position. Experts' ratings of both avatar versions fall with sentence length ($\rho = -.42$ for our avatar and $-.41$ for the gold-gloss avatar, $p = .04$; short sentences 2.55 vs. long 1.86), while their ratings of the real signer do not. Deaf ratings of our avatar do not depend on length ($\rho = -.11$). Deaf ratings declined slightly over the session (our avatar $\rho = -.27$, $p = .03$; real signer $\rho = -.24$, $p = .06$), while experts rated later avatar videos higher ($\rho = .28$, $p = .01$).

Reference translations. Experts commented that some TİD Sözlük sentence videos are free translations of the Turkish sentence rather than close translations (e.g., items 1003-03 and 1546-01). This also limits gloss-level metrics computed against these references (Section 5.2).

		Mean of three scales				Blind translations	
Expert	Background	Our avatar	Gold gloss	Real	Und. ≥ 3	Sem.	chrF
E1	Deaf, TİD from birth, researcher	3.12	3.23	4.99	65%	0.82	46.7
E2	hearing, researcher, 15 y	1.77	1.59	5.00	12%	0.64	24.8
E3	hearing, researcher, 11 y	1.84	2.16	4.89	16%	0.65	27.6
E4	hearing, teacher, 14 y	2.89	3.11	4.94	50%	0.72	37.8
All		2.40	2.52	4.96	32%	0.72	34.6

Table 11: Expert raters. Years are years of TİD use. Und. ≥ 3: share of our avatar videos rated 3 or higher for understanding. Sem.: semantic similarity to the source sentence (model of Table 1).

I Pose Metrics with WiLoR Hands

All pose metrics in the paper use MediaPipe Holistic landmarks. To check whether the conclusions depend on the hand estimator, we ran WiLoR (Potamias et al., 2025) (WiLoR-mini, commit ebec42f, float32, detection threshold 0.3) on the same 371 test videos (71,882 frames). WiLoR fits the MANO hand model, whereas MediaPipe predicts 21 keypoints directly; frame counts, hand labels and keypoint order agree between the two outputs. Since no hand ground truth exists for these videos, we report coverage and agreement rather than accuracy. Distances are given in shoulder widths (SW, the median inter-shoulder distance per video), with 95% CIs bootstrapped over videos.

Coverage and agreement. WiLoR detected the right hand in 88.7% of frames (MediaPipe 82.4%) and the left hand in 64.9% (62.8%), and missed a hand whose wrist lay inside the image in 12.3% of frames (17.0%). Across both hands, 71.4% of hand-frames are detected by both estimators, 5.4% by WiLoR only, 1.2% by MediaPipe only and 22.0% by neither; 99.5% of the WiLoR-only detections lie within 0.5 SW of the body-pose wrist. Where both estimators detect a hand, the mean 2D joint distance is 0.038 SW (95% CI 0.037 to 0.039) and exceeds 0.25 SW in 0.5% of frames; palm normals differ by a median of 19.1 degrees (IQR 12.0 to 28.1).

Rescored pose metrics. We replaced the hands in the reference poses with WiLoR hands and rescored the stored generated sequences without regenerating them. As the generated hands still come from MediaPipe, the ablation effects, which compare two generated versions against the same reference, are more informative than the absolute errors. Table 12 shows that Hands errors fall by 3 to 6% for every system, the system ranking is unchanged, and both ablation effects keep their direction and significance. The agreement effect halves for Gemma 3 27B (0.0013 to 0.0007), is unchanged for the other three systems and rises slightly for the ground-truth stitch (0.0019 to 0.0022); the context-aware resolution effect is 13 to 25% smaller for every system but remains significant and larger than the agreement effect. We therefore report these effects as ranges.

	DTW Hands		Agreement		Context-aware	
System	MP	WiLoR	MP	WiLoR	MP	WiLoR
Gemma 3 27B	0.1669	0.1581	0.0013**	0.0007*	0.0047***	0.0036***
Qwen 3 32B	0.1679	0.1590	0.0003	0.0004	0.0041***	0.0035***
GPT-5.4	0.1684	0.1592	0.0012**	0.0012**	0.0038***	0.0031***
Gemini 3 Flash	0.1689	0.1594	0.0018***	0.0017***	0.0041***	0.0036***
Gold stitch	0.1599	0.1514	0.0019***	0.0022***	0.0051***	0.0038***

Table 12: DTW Hands error of the full pipeline against the real video with MediaPipe (MP) or WiLoR reference hands, and the increase in DTW Hands when the agreement or the context-aware resolution module is disabled. Holm-corrected paired Wilcoxon tests across the five systems: * $p < .05$, ** $p < .01$, *** $p < .001$. Agreement: 169 sentences with an agreeing verb. Context-aware resolution: sentences where sense disambiguation or semantic nearest-neighbor substitution was used (261 to 281); disabling it turns off sense disambiguation, nearest-neighbor substitution and fingerspelling. Gemma 3 27B, Qwen 3 32B fine-tuned; GPT-5.4, Gemini 3 Flash ten-shot.